Amirhossein Abaskohi

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Research Interests

My research focuses on reliable and efficient language-model agents for enterprise systems. I study how agents gather evidence, reason over procedural knowledge, and plan across complex workflows, as well as how diversity in model outputs affects exploration and reasoning.

- **Deep research agents:** Benchmarking, evidence-grounded reasoning, and predicting and repairing premature commitments.
- **Enterprise workflows and world models:** Representing procedural knowledge, learning probabilistic workflow models, and supporting planning across organizational processes.
- **Diversity in language models:** Output distributions, homogeneity, and diverse exploration in open-ended tasks.
- **Efficient reasoning and retrieval:** Retrieval-augmented generation, long multimodal document understanding, computer-use agents, and inference-time adaptation.

Education

University of British Columbia Sep 2025 – Present
Ph.D. in Computer Science
Supervisors: Giuseppe Carenini and Peter West.
GPA: A+.

University of British Columbia Sep 2023 – Aug 2025
M.Sc. in Computer Science
Research advisors: Giuseppe Carenini and Issam H. Laradji.
Thesis: *Multimodal Understanding of Long Documents: From Topic Modeling to Question Answering*.
GPA: A+.

University of Tehran Sep 2018 – Apr 2023
B.Sc. in Computer Engineering
CGPA: 3.8/4.0; final two years: 4.0/4.0. Ranked in the top 10% of the class.

Research Experience

Microsoft Research Redmond (Remote) — Research Intern Jun 2026 – August 2026

Microsoft Research Asia, Vancouver — Research Intern Jan 2026 – March 2026

ServiceNow Research — Research Intern 2024 – 2025

Peer-Reviewed Publications

2026

[1] **CLARUS: Hierarchical Hypergraph Modeling for Unsupervised Discharge Summarization**
G. Cimino, M. R. I. Hossain, L. De Santis, **A. Abaskohi**, R. T. Ng, G. Carenini, V. Deufemia
EMNLP 2026. [Paper]

[2] **MCompassRAG: Topic Metadata as a Semantic Compass for Paragraph-Level Retrieval**
A. Abaskohi, R. Li, G. Cimino, P. West, G. Carenini, I. H. Laradji
Findings of EMNLP 2026. [Paper]

[3] **SproutRAG: Attention-Guided Tree Search with Progressive Embeddings for Long-Document RAG**
A. Abaskohi, I. H. Laradji, P. West, G. Carenini
Findings of EMNLP 2026. [Paper]

[4] **ReVision: Scaling Computer-Use Agents via Temporal Visual Redundancy Reduction**
A. Abaskohi, Y. He, P. West, G. Carenini, P. Chawla, V. Vineet
Findings of EMNLP 2026. [Paper]

[5] **DeepRewind: Predicting and Repairing Premature Commitments in Deep Research Agents**
A. Abaskohi, A. Dabiriaghdam, L. Wang, P. West, G. Carenini
REALM Workshop at EMNLP 2026. [Paper]

[6] **SEDIMA: Cross-Run Hierarchical Insight Memory for Evolutionary Search Agents**
A. Abaskohi, M. Mostajabdaveh, Z. Zhou
REALM Workshop at EMNLP 2026. [Paper] **Spotlight**

[7] **Improving Topic Modeling by Distilling Soft Labels from Language Models**
R. Li, **A. Abaskohi**, C. Li, G. Murray, G. Carenini
ICML 2026. [Paper]

[8] **DRBench: A Realistic Benchmark for Enterprise Deep Research**
A. Abaskohi, T. Chen, M. Muñoz-Mármol, C. Fox, A. V. Ramesh, É. Marcotte, X. H. Lù, N. Chapados, S. Gella, P. West, G. Carenini, C. Pal, A. Drouin, I. H. Laradji
ICLR 2026. [Paper]

[9] **StarFlow: Generating Structured Workflow Outputs From Sketch Images**
P. Bechard, C. Wang, **A. Abaskohi**, J. Rodriguez, C. Pal, D. Vazquez, S. Gella, S. Rajeswar, P. Taslakian
EACL 2026. [Paper]

2025

[10] **CEMTM: Contextual Embedding-based Multimodal Topic Modeling**
A. Abaskohi, R. Li, C. Li, S. Joty, G. Carenini
EMNLP 2025. [Paper]

[11] **ChartGaze: Enhancing Chart Understanding in LVLMs with Eye-Tracking Guided Attention Refinement**
A. Salamatian, **A. Abaskohi**, W.-C. Fan, M. R. I. Hossain, L. Sigal, G. Carenini
EMNLP 2025. [Paper]

[12] **FM^{2}DS: Few-Shot Multimodal Multihop Data Synthesis with Knowledge Distillation for Question Answering**
A. Abaskohi, S. Gella, G. Carenini, I. H. Laradji
Findings of EMNLP 2025. [Paper]

[13] **AgentAda: Skill-Adaptive Data Analytics for Tailored Insight Discovery**
A. Abaskohi, A. V. Ramesh, S. Nanisetty, C. Goel, D. Vazquez, C. Pal, S. Gella, G. Carenini, I. H. Laradji
REALM Workshop at ACL 2025. [Paper] **Oral, Outstanding Paper Award**

[14] **BigDocs: A Permissively-Licensed Dataset for Training Vision-Language Models on Document and Code Tasks**
J. A. Rodriguez, X. Jian, S. S. Panigrahi, T. Zhang, A. Feizi, A. Puri, A. Kalkunte, F. Savard, **A. Abaskohi** (equal second-author contribution), et al.
ICLR 2025; also RBFM Workshop at NeurIPS 2024. [Paper]

[15] **InsightBench: Evaluating Business Analytics Agents Through Multi-Step Insight Generation**
G. Sahu, A. Puri, J. Rodriguez, **A. Abaskohi**, M. Chegini, A. Drouin, P. Taslakian, V. Zantedeschi, A. Lacoste, D. Vazquez, N. Chapados, C. Pal, S. Rajeswar Mudumba, I. H. Laradji
ICLR 2025. [Paper]

2024

[16] **Benchmarking Large Language Models for Persian: A Preliminary Study Focusing on ChatGPT**
A. Abaskohi, S. Baruni, M. Masoudi, N. Abbasi, M. H. Babalou, A. Edalat, S. Kamahi, S. Mahdizadeh Sani, N. Naghavian, D. Namazifard, P. Sadeghi, Y. Yaghoobzadeh
LREC-COLING 2024. [Paper]

[17] **BCAmirs at SemEval-2024 Task 4: Beyond Words: A Multimodal and Multilingual Exploration of Persuasion in Memes**
A. Abaskohi[*], A. Dabiriaghdam[*], L. Wang, G. Carenini
SemEval Workshop at NAACL 2024. [Paper]

[18] **uTeBC-NLP at SemEval-2024 Task 9: Can LLMs be Lateral Thinkers?**
P. Sadeghi[*], **A. Abaskohi**[*], Y. Yaghoobzadeh
SemEval Workshop at NAACL 2024. [Paper]

2023

[19] **LM-CPPF: Paraphrasing-Guided Data Augmentation for Contrastive Prompt-Based Few-Shot Fine-Tuning**
A. Abaskohi, S. Rothe, Y. Yaghoobzadeh
ACL 2023. [Paper]

[20] **PEACH: Pre-Training Sequence-to-Sequence Multilingual Models for Translation with Semi-Supervised Pseudo-Parallel Document Generation**
A. Abaskohi[*], A. Salemi[*], S. Tavakoli, Y. Yaghoobzadeh, A. Shakery
LoResMT Workshop at EACL 2023. [Paper]

[21] **A Large-Scale Analysis of Persian Tweets Regarding Covid-19 Vaccination**
T. ShabaniMirzaei, H. Chamani, **A. Abaskohi**, Z. Sourati Hassan Zadeh, B. Bahrak
Social Network Analysis and Mining, 2023. [Paper]

2022

[22] **UTNLP at SemEval-2022 Task 6: A Comparative Analysis of Sarcasm Detection Using Generative-based and Mutation-based Data Augmentation**
A. Abaskohi, A. Rasouli*, T. Zeraati*, B. Bahrak
SemEval Workshop at NAACL 2022. [Paper]

Preprints

[P1] **Dual-Confidence Contrastive Decoding for Retrieval-Augmented Generation**
R. Li, Md. T. I. Khondaker, **A. Abaskohi**, G. Murray, G. Carenini, I. H. Laradji
arXiv:2607.00570, 2026. [Paper]

[P2] **SeKV: Resolution-Adaptive KV Cache with Hierarchical Semantic Memory for Long-Context LLM Inference**
A. Abaskohi, G. Carenini, P. West, Y. He
arXiv:2606.31145, 2026. [Paper]

[P3] **UnpredictaBench: A Benchmark for Evaluating Distributional Randomness in LLMs**
A. Abaskohi*, A. Dabiriaghdam*, L. Luo, E. D. Wen, L. Wang, G. Carenini, P. West
arXiv:2606.06622, 2026. [Paper]

[P4] **Persian Emotion Detection using ParsBERT and Imbalanced Data Handling Approaches**
A. Abaskohi, N. Sabri, B. Bahrak
arXiv:2211.08029, 2022. [Paper]

[P5] **Automatic Speech Recognition for Speech Assessment of Persian Preschool Children**
A. Abaskohi, F. Mortazavi, H. Moradi
arXiv:2203.12886, 2022. [Paper]

Honors and Awards

Spotlight Paper, SEDIMA, REALM at EMNLP	2026
Outstanding Paper Award, AgentAda, REALM at ACL	2025
President's Academic Excellence Initiative PhD Award, UBC	2025
Graduate Support Initiative Award, UBC	2025
Mitacs Fellowship, Project IT41557	2024
International Tuition Award, UBC	2023
CS Strife Award, UBC	2023
Best Undergraduate Project Award, University of Tehran	2022
Supporter Foundation of University of Tehran Scholarship	2020

Invited Talks

Latest Developments in Efficient Retrieval-Augmented Generation Aug 2026

VAM! Talks: Frontier AI Safety & Efficient RAG, Vancouver AI Meetup, Vancouver, BC.

DRBench: A Realistic Benchmark for Enterprise Deep Research Aug 2025

Friday Breakfast Talks, Microsoft Research India (Host: Niket Tandon).

Teaching Experience

Teaching Assistant, University of British Columbia

CPSC 423: Natural Language Processing Fall 2026

CPSC 422: Intelligent Systems Spring 2025, 2026

CPSC 213: Introduction to Computer Systems Fall 2025

CPSC 322: Introduction to Artificial Intelligence Fall 2023; Spring, Fall 2024

Academic Service and Outreach

Publicity Chair, Future of NLP Workshop Oct – Dec 2024

AI/ML Mentor, Summer School Summer 2021

Technical Content Creator: Medium and YouTube Jul 2021 – Present

Technical Skills

LLM Training & Post-training: PyTorch, Hugging Face (Transformers, PEFT, TRL, Accelerate), DeepSpeed, FSDP; SFT, LoRA/QLoRA, preference and RL-based post-training (DPO, GRPO), activation steering and representation engineering.

Inference & Efficiency: vLLM, SGLang, FlashAttention, CUDA; KV-cache compression, long-context inference, quantization.

Agents & Retrieval: deep research agents, computer-use agents, multi-agent systems, tool use and MCP, LangGraph, Playwright; retrieval-augmented generation with FAISS, Elasticsearch, and dense/hybrid retrievers.

Evaluation: benchmark design, LLM-as-a-judge, distributional evaluation, lm-evaluation-harness, Weights & Biases.

Programming: Python, C/C++, Java, C#, SQL, JavaScript, Bash.

Infrastructure: Git, Docker, Slurm/HPC clusters, AWS, Linux, PostgreSQL, MongoDB, Spark.